

1 Goods Receipt MES --> ERP

Overview

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The interface transfers all output batches generated from production orders in the MES for inventory-managed material maintained in the ERP system to the ERP system.

The interface transfers the tree of origin and the user-defined fields (of the producing operation) for each lot (= batch).

The MES provides the data at regular intervals to the ERP system. The IDoc is of the type ZWEI02. This leads to the following specification:

Message type: **ZWEI**

File name (for file transfers) **Z2WEI000X000**

File extension (for file transfers): Depending on the configuration (by default ".dat")

IDOC type (with tRFC communication): **ZWEI02**

Segments:	Z2WEI000X000 (goods receipts)	1 – n
	Z2CNRATT_C000X000 (alphanumeric batch attributes part 1)	
	0 – 1	
	Z2CNRATT_C001X000 (alphanumeric batch attributes part 2)	
	0 – 1	
	Z2CNRATT_N000X000 (numeric batch attributes) - OBSOLETE ¹	0 – 1
	Z2CNRATT_N001X000 (numeric batch attributes) ²	0 – 1
	Z2CNRBAUM000X000 (optional: tree of generation)	0 – n
	Z2TOLO000X000 (optional: sub-batches)	0 – n
	Z2CNR_USRFLD000X000 (optional: user fields) ³	0 – 1

¹You should no longer use this segment for new installations. Use the segment Z2CNRATT_N001X000 instead. The segment Z2CNRATT_N000X000 is still available (backwards compatible) but will no longer be maintained. Both segments have different field lengths for their decimal fields.

² Please note the information on the activation in section Numeric attributes (Z2CNRATT_N001X000).

³ Please note the information on the activation in section User fields (Z2CNR_USRFLD000X000).

Note the following conventions if SAP is connected:



Create SAP segment names according to the pattern Z1<segment name> in order to generate the above-mentioned segment names in SAP. Versioning in SAP outbound processing then generates segment names in the form Z2<Segment name><Version>.

Example: the created segment name **Z1WEI000X** becomes **Z2WEI000X000**

Movement types

Goods receipts are transferred from the MES to the ERP system using the IDoc and message types described below. The movement type specifies the action (goods receipt, stock transfer, etc.) that must be triggered in the ERP system.

The following table describes which movement type is used to transfer goods receipt postings to the ERP system:

Movement type	Description/ usage
101	Standard goods receipt from production order: Goods receipt for batches where the option "Transfer to interface" is set for the material type.
102	Cancellation of goods receipt
525	Goods receipt for batches that are blocked in the MES.
531	Goods receipt for waste batches
532	Cancellation of goods receipt for waste batches and batches

Goods receipt (Z2WEI000X000)

The data record described below transfers the goods receipt. The detailed information:

- user-specific fields
- tree of generation
- sub-batches for pallets

is transferred in sub-segments.

Field	T	L	D	Description
WERK	CHAR	4		Plant; stored in HYDRA (fixed)
BEWART	CHAR	3		Movement type: <i>see table above</i>
GRUND	NUM	4		e.g. blocking reason for BEWART 525

Field	T	L	D	Description
ZLO	CHAR	12		Target material buffer
LGORT	CHAR	20		ERP storage location Storage location stored to the target material buffer.
LGPZ	CHAR	20		ERP storage bin
CHARGE	CHAR	10		ERP batch number
MATNR	CHAR	40		Material number
MATTYP	CHAR	10		Material type in HYDRA
MENGE	QUAN	13	3	Quantity (primary quantity of the producing operation)
MENGE_EINH	CHAR	3		Quantity unit (primary quantity of the producing operation) Unit for the quantity in MENGE
ANR	CHAR	40		HYDRA order number = combined order/ operation number The exact length that is uploaded/confirmed depends on how the lengths are configured for the order or operation in the HYDRA basic parameter settings. Used for ERP inbound processing if SAP is not in use.
SAP_AUNR	12			SAP order number Used for ERP inbound processing if SAP is in use.
SAP_AFOLG	6			SAP sequence number Used for ERP inbound processing if SAP is in use.
SAP_VORNR	4			SAP operation number Used for ERP inbound processing if SAP is in use.
SAP_UVGNR	4			SAP sub-operation number Used for ERP inbound processing if SAP is in use.
KZSOBEST	CHAR	1		Not used, populated with " " (blanks).
KZ_ABZWEIG	CHAR	1		Indicates if the material derives from the planned deducted material: Comes from the OP data record and has the following meaning: "M" Master OP of planned deducted material "K" Sub-OP of planned deducted material " " other
END_LIEF	CHAR	1		Indicates if this is the last batch of the operation. "J" last batch "N" other The value "J" is set if the last output batch is completed when logging off the OP.
INDEX	NUMC	4		Index (counter) of the batches within a production order.
HY_LOSNR	CHAR	20		HYDRA batch number
HY_DLLNR	CHAR	20		HYDRA throughput batch number
Z_MENGE	QUAN	13	3	Not used; by default: 0
Z_MEINH	CHAR	3		Not used; by default: " "
ARBPL	CHAR	8		Producing HYDRA machine
ANZ_TR	NUMC	8		Pallet (package): Number of individual batches for the pallet

Field	T	L	D	Description
LOSHINWEIS	CHAR	20		Entered information on batch
LST01	QUAN	13	3	Activity 1
LST01_EINH	CHAR	3		Unit of activity 1
LST02	QUAN	13	3	Activity 2
LST02_EINH	CHAR	3		Unit of activity 2
LST03	QUAN	13	3	Activity 3
LST03_EINH	CHAR	3		Unit of activity 3
LST04	QUAN	13	3	Activity 4
LST04_EINH	CHAR	3		Unit of activity 4
LST05	QUAN	13	3	Activity 5
LST05_EINH	CHAR	3		Unit of activity 5
LST06	QUAN	13	3	Activity 6
LST06_EINH	CHAR	3		Unit of activity 6
VVDAT	DATE	8		Availability date (MPL-MMO) If MPL-MMO is not used, you should set the value to the current point in time.
VVZEI	TIME	6		Availability time (MPL-MMO) If MPL-MMO is not used, you should set the value to the current point in time.
WDAT	DATE	8		Warning date (MPL-MMO) If MPL-MMO is not used, you should set the value to 31.12.9999.
WZEI	TIME	6		Warning time (MPL-MMO) If MPL-MMO is not used, you should set the value to 23:59:59.
VFDAT	DATE	8		Expiry date (MPL-MMO) If MPL-MMO is not used, you should set the value to 31.12.9999.
VFZEI	TIME	6		Expiry time (MPL-MMO) If MPL-MMO is not used, you should set the value to 23:59:59.
KLASSE	CHAR	1		Batch class "G" Yield "A" Scrap/ waste As of MPL 8.2 the following, additional indicators are available: "O" Open quantity / problem quantity "N" Rework
STATUS	CHAR	1		Batch status to be set F Free/available S Blocked
MATST	CHAR	1		Material status
QST	CHAR	1		Quality status
QSTMANU	CHAR	1		Manual quality status
TST	CHAR	1		Transport status

Field	T	L	D	Description
PERSNO	NUM	8		Personnel number
TPE	CHAR	10		Transport unit
CNR_ALT1	CHAR	20		Alternative batch number 1
CNR_ALT2	CHAR	20		Alternative batch number 2
CNR_ALT3	CHAR	20		Alternative batch number 3
CNR_ALT4	CHAR	20		Alternative batch number 4
CNR_ALT5	CHAR	40		Alternative batch number 5
EXTCNR	CHAR	20		External batch number (e.g. batch number)
MCNR	CHAR	20		Merged batch number
ATTR1	NUM	8		Direct batch attribute 1
ATTR2	NUM	8		Direct batch attribute 2
ATTR3	NUM	8		Direct batch attribute 3
ATTR4	DEC	13	3	Direct batch attribute 4
ATTR5	DEC	13	3	Direct batch attribute 5
ATTR6	DEC	13	3	Direct batch attribute 6
ATTR7	CHAR	4		Direct batch attribute 7
ATTR8	CHAR	10		Direct batch attribute 8
ATTR9	CHAR	10		Direct batch attribute 9
ATTR10	CHAR	20		Direct batch attribute 10
CHARGE_LONG	CHAR	20		ERP batch number (long) Available from MPL 8.2 on - please also see the following information on field CHARGE_LONG
MSL_VFDATE	DATE	8		MSL expiry date This field is only available if the database patch "dbp_mpl_mslmonitoring.hsc" is executed.
MSL_VFTIME	TIME	6		MSL expiry time This field is only available if the database patch "dbp_mpl_mslmonitoring.hsc" is executed.
MSL_PERIOD	NUMC	8		MSL term This field is only available if the database patch "dbp_mpl_mslmonitoring.hsc" is executed.

Please note: Batch-related fields are only populated if the movements refer to batches.

Information on the fields CHARGE / CHARGE_LONG

The field CHARGE_LONG is only available as of MPL 8.2.

If MPL 8.2 is used, the fields CHARGE and CHARGE_LONG are populated as follows:

CHARGE includes the ERP batch number with the characters 1-10.

CHARGE_LONG includes the ERP batch number with the characters 1-20.

You should use the value of the CHARGE_LONG field for new installations.

Alphanumeric attributes (Z2CNRATT_C000X000)

The following segment transfers the (first 20) alphanumeric batch attributes.

Field	T	L	D	Description
HY_LOSNR	CHAR	20		HYDRA batch number
ATTRIB_101	CHAR	40		Alphanumeric batch attribute
ATTRIB_102	CHAR	40		...
ATTRIB_103	CHAR	40		...
ATTRIB_104	CHAR	40		...
ATTRIB_105	CHAR	40		...
ATTRIB_106	CHAR	40		...
ATTRIB_107	CHAR	40		...
ATTRIB_108	CHAR	40		...
ATTRIB_109	CHAR	40		...
ATTRIB_110	CHAR	40		...
ATTRIB_111	CHAR	40		...
ATTRIB_112	CHAR	40		...
ATTRIB_113	CHAR	40		...
ATTRIB_114	CHAR	40		...
ATTRIB_115	CHAR	40		...
ATTRIB_116	CHAR	40		...
ATTRIB_117	CHAR	40		...
ATTRIB_118	CHAR	40		...
ATTRIB_119	CHAR	40		...
ATTRIB_120	CHAR	40		...

Alphanumeric attributes (Z2CNRATT_C001X000)

The following segment transfers the (first 20) alphanumeric batch attributes.

Field	T	L	D	Description
HY_LOSNR	CHAR	20		HYDRA batch number
ATTRIB_121	CHAR	40		Alphanumeric batch attribute
ATTRIB_122	CHAR	40		...
ATTRIB_123	CHAR	40		...
ATTRIB_124	CHAR	40		...
ATTRIB_125	CHAR	40		...
ATTRIB_126	CHAR	40		...
ATTRIB_127	CHAR	40		...
ATTRIB_128	CHAR	40		...
ATTRIB_129	CHAR	40		...
ATTRIB_130	CHAR	40		...
ATTRIB_131	CHAR	40		...
ATTRIB_132	CHAR	40		...
ATTRIB_133	CHAR	40		...
ATTRIB_134	CHAR	40		...
ATTRIB_135	CHAR	40		...
ATTRIB_136	CHAR	40		...
ATTRIB_137	CHAR	40		...
ATTRIB_138	CHAR	40		...
ATTRIB_139	CHAR	40		...
ATTRIB_140	CHAR	40		...

Numeric attributes (Z2CNRATT_N000X000) - OBSOLETE

The following segment transfers the numeric batch attributes.

Field	T	L	D	Description
HY_LOSNR	CHAR	20		HYDRA batch number
ATTRIB_201	NUMC	8		Integer, numeric batch attribute
ATTRIB_202	NUMC	8		...
ATTRIB_203	NUMC	8		...
ATTRIB_204	NUMC	8		...
ATTRIB_205	NUMC	8		...
ATTRIB_206	NUMC	8		...
ATTRIB_207	NUMC	8		...
ATTRIB_208	NUMC	8		...
ATTRIB_209	NUMC	8		...
ATTRIB_210	NUMC	8		...
ATTRIB_211	NUMC	8		...
ATTRIB_212	NUMC	8		...
ATTRIB_213	NUMC	8		...
ATTRIB_214	NUMC	8		...
ATTRIB_215	NUMC	8		...
ATTRIB_216	NUMC	8		...
ATTRIB_217	NUMC	8		...
ATTRIB_218	NUMC	8		...
ATTRIB_219	NUMC	8		...
ATTRIB_220	NUMC	8		...
ATTRIB_301	DEC	10	3	Decimal, numeric batch attribute
ATTRIB_302	DEC	10	3	...
ATTRIB_303	DEC	10	3	...
ATTRIB_304	DEC	10	3	...
ATTRIB_305	DEC	10	3	...
ATTRIB_306	DEC	10	3	...
ATTRIB_307	DEC	10	3	...
ATTRIB_308	DEC	10	3	...
ATTRIB_309	DEC	10	3	...
ATTRIB_310	DEC	10	3	...
ATTRIB_311	DEC	10	3	...
ATTRIB_312	DEC	10	3	...
ATTRIB_313	DEC	10	3	...
ATTRIB_314	DEC	10	3	...
ATTRIB_315	DEC	10	3	...
ATTRIB_316	DEC	10	3	...
ATTRIB_317	DEC	10	3	...
ATTRIB_318	DEC	10	3	...
ATTRIB_319	DEC	10	3	...
ATTRIB_320	DEC	10	3	...

Numeric attributes (Z2CNRATT_N001X000)

As of version 1.72946 (2015/SP7) of the script *mle_rckmestyp_zwei_out.hsc*, you can use the segment Z2CNRATT_N001X000 to upload/confirm the numeric attributes of a batch.

Depending on the product version in use, you might have to activate the segment manually:

MPL/TRT 8.1	MPL/TRT 8.2
You have to enable uploads via the segment Z2CNRATT_N001X000 manually.	For new installations after SP7/2015 the segment is used by default. You have to enable the segment manually for installations prior to that date.

You have to enable the transfer of the segment manually in the [HYDRA INI configuration](#).

Data is read from the HYDRA table *los_bestand*

Field	T	L	D	Description
HY_LOSNR	CHAR	20		HYDRA batch number
ATTRIB_201	NUMC	8		Integer, numeric batch attribute
ATTRIB_202	NUMC	8		...
ATTRIB_203	NUMC	8		...
ATTRIB_204	NUMC	8		...
ATTRIB_205	NUMC	8		...
ATTRIB_206	NUMC	8		...
ATTRIB_207	NUMC	8		...
ATTRIB_208	NUMC	8		...
ATTRIB_209	NUMC	8		...
ATTRIB_210	NUMC	8		...
ATTRIB_211	NUMC	8		...
ATTRIB_212	NUMC	8		...
ATTRIB_213	NUMC	8		...
ATTRIB_214	NUMC	8		...
ATTRIB_215	NUMC	8		...
ATTRIB_216	NUMC	8		...
ATTRIB_217	NUMC	8		...
ATTRIB_218	NUMC	8		...
ATTRIB_219	NUMC	8		...
ATTRIB_220	NUMC	8		...

Field	T	L	D	Description
ATTRIB_301	DEC	13	3	Decimal, numeric batch attribute
ATTRIB_302	DEC	13	3	...
ATTRIB_303	DEC	13	3	...
ATTRIB_304	DEC	13	3	...
ATTRIB_305	DEC	13	3	...
ATTRIB_306	DEC	13	3	...
ATTRIB_307	DEC	13	3	...
ATTRIB_308	DEC	13	3	...
ATTRIB_309	DEC	13	3	...
ATTRIB_310	DEC	13	3	...
ATTRIB_311	DEC	13	3	...
ATTRIB_312	DEC	13	3	...
ATTRIB_313	DEC	13	3	...
ATTRIB_314	DEC	10	3	...
ATTRIB_315	DEC	10	3	...
ATTRIB_316	DEC	10	3	...
ATTRIB_317	DEC	10	3	...
ATTRIB_318	DEC	10	3	...
ATTRIB_319	DEC	10	3	...
ATTRIB_320	DEC	10	3	...

Tree of generation (Z2CNRBAUM000X000)

The following data record is part of the tree of generation of a batch. These are "OPTIONAL" segments, i.e. they are only included in the IDoc if they actually exist.

Data is read from the HYDRA table **LOS_ZUORDNUNGEN**.

Field	T	L	D	Description
HY_ALOSNR lz.al_nr	CHAR	20		HYDRA batch number (output batch)
A_CHARGE lz.al_charge	CHAR	10		ERP batch number of the output batch ... populated if the batch is an ERP batch.
A_MATNR lz.al_matnr	CHAR	40		Material number of the output batch Known in the ERP system for inventory-managed materials.
HY_ELOSNR	CHAR	20		HYDRA batch number (input batch)
E_CHARGE	CHAR	10		ERP batch number of the input batch ... populated if the batch is an ERP batch.
E_MATNR	CHAR	40		Material number of the input batch Known in the ERP system for inventory-managed materials.
E_POS	CHAR	10		BOM item of the input batch
ANR	CHAR	40		HYDRA production order producing the output batch.

Field	T	L	D	Description
ARBPL	CHAR	8		Machine where the output batch was generated.
DATUM	DATE	8		Creation date of the output batch
ZEIT	TIME	6		Creation time of the output batch

Note

You should manage this information in a user-specific table "ZCNRBAUM" in the ERP system. The structure of this table should be identical to that of the segment.

Sub-batches (Z2TOLO000X000)

For pallets: The described sub-segment transfers the included sub-batches that are assigned to the pallets. These are "OPTIONAL" segments, i.e. they are only included in the IDoc if they actually exist.

Data is read from the HYDRA table **ZTOLO**.

Field	T	L	D	Description
HY_LOSNR	CHAR	20		HYDRA batch number of individual batch
MATNR	CHAR	40		HYDRA material number
MATTYP	CHAR	10		HYDRA material type
MATTXT	CHAR	40		HYDRA material name
MENGE	QUAN	13	3	Quantity of individual batch
MENGE_EINH	CHAR	3		Quantity unit
LST01	QUAN	13	3	Activity 1
LST01_EINH	CHAR	3		Unit of activity 1
LST02	QUAN	13	3	Activity 2
LST02_EINH	CHAR	3		Unit of activity 2
LST03	QUAN	13	3	Activity 3
LST03_EINH	CHAR	3		Unit of activity 3
LST04	QUAN	13	3	Activity 4
LST04_EINH	CHAR	3		Unit of activity 4
LST05	QUAN	13	3	Activity 5
LST05_EINH	CHAR	3		Unit of activity 5
LST06	QUAN	13	3	Activity 6
LST06_EINH	CHAR	3		Unit of activity 6

User fields (Z2CNR_USRFLD000X000)

You can transfer/upload the user fields of a batch as of version 1.8 of the script *mle_rckmestyp_zwei_out.hsc*. Data is transferred in the segment Z2CNR_USRFLD000X000.

You have to enable the transfer of the segment manually in the [HYDRA INI configuration](#).

Data is read from the HYDRA table *los_bestand*

Field	V	T	L	D	Description
HY_LOSNR	S	CHAR	20		HYDRA batch number
FILLER	S	CHAR	20		Placeholder – internal usage
USRFLD	S	CHAR	8		User field key
FU:1	K	DATE	8		User field 1
FU:2	K	DATE	8		User field 2
FU:3	K	DATE	8		User field 3
FU:4	K	DATE	8		User field 4
FU:5	K	DATE	8		User field 5
FU:6	K	DATE	8		User field 6
FU:7	K	NUM	8		User field 7
FU:8	K	NUM	8		User field 8
FU:9	K	NUM	8		User field 9
FU:10	K	NUM	8		User field 10
FU:11	K	NUM	8		User field 11
FU:12	K	NUM	8		User field 12
FU:13	K	NUM	8		User field 13
FU:14	K	NUM	8		User field 14
FU:15	K	NUM	8		User field 15
FU:16	K	NUM	8		User field 16
FU:17	K	NUM	8		User field 17
FU:18	K	NUM	8		User field 18
FU:19	K	NUM	8		User field 19
FU:20	K	NUM	8		User field 20
FU:21	K	NUM	8		User field 21
FU:22	K	NUM	8		User field 22
FU:23	K	DEC	13	3	User field 23
FU:24	K	DEC	13	3	User field 24
FU:25	K	DEC	13	3	User field 25
FU:26	K	DEC	13	3	User field 26
FU:27	K	DEC	13	3	User field 27
FU:28	K	DEC	13	3	User field 28
FU:29	K	CHAR	1		User field 29
FU:30	K	CHAR	1		User field 30
FU:31	K	CHAR	1		User field 31
FU:32	K	CHAR	1		User field 32
FU:33	K	CHAR	1		User field 33
FU:34	K	CHAR	1		User field 34

Field	V	T	L	D	Description
FU:35	K	CHAR	1		User field 35
FU:36	K	CHAR	1		User field 36
FU:37	K	CHAR	1		User field 37
FU:38	K	CHAR	1		User field 38
FU:39	K	CHAR	1		User field 39
FU:40	K	CHAR	1		User field 40
FU:41	K	CHAR	1		User field 41
FU:42	K	CHAR	1		User field 42
FU:43	K	CHAR	1		User field 43
FU:44	K	CHAR	1		User field 44
FU:45	K	CHAR	10		User field 45
FU:46	K	CHAR	10		User field 46
FU:47	K	CHAR	10		User field 47
FU:48	K	CHAR	10		User field 48
FU:49	K	CHAR	10		User field 49
FU:50	K	CHAR	10		User field 50
FU:51	K	CHAR	20		User field 51
FU:52	K	CHAR	20		User field 52
FU:53	K	CHAR	20		User field 53
FU:54	K	CHAR	20		User field 54
FU:55	K	CHAR	20		User field 55
FU:56	K	CHAR	20		User field 56
FU:57	K	CHAR	20		User field 57
FU:58	K	CHAR	20		User field 58
FU:59	K	CHAR	20		User field 59
FU:60	K	CHAR	20		User field 60
FU:61	K	CHAR	20		User field 61
FU:62	K	CHAR	20		User field 62
FU:63	K	CHAR	20		User field 63
FU:64	K	CHAR	20		User field 64
FU:65	K	CHAR	40		User field 65
FU:66	K	CHAR	40		User field 66